

REBUILD PLAN · 18 AUG 2026

# Sedna Ops — Gap Analysis & Build Plan

28 identified gaps across 5 categories, folded into 7 sequential phases. From static prototype to a persistent, enforcing, customer-centric operations platform with an AI intelligence layer.

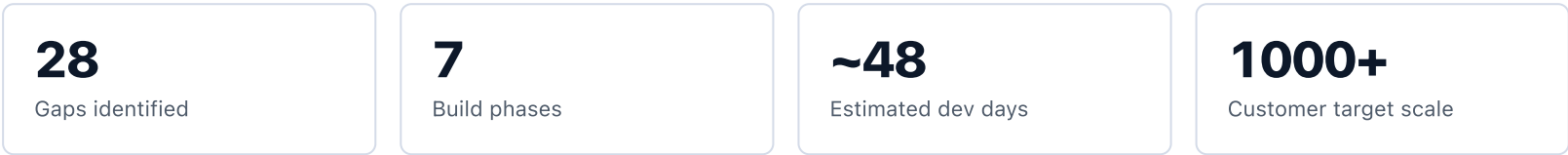
Vue 3 + Vite + Pinia

FastAPI + SQLAlchemy 2.0 + Alembic

PostgreSQL 16

Docker Compose · 4 services

Haiku 4.5 + Sonnet 5



GAP INDEX — 28 GAPS ACROSS 5 CATEGORIES

|    |                                                                                                |                           |        |
|----|------------------------------------------------------------------------------------------------|---------------------------|--------|
| A  | Process Enforcement                                                                            | CAN CAUSE INCIDENTS TODAY | 8 gaps |
| A1 | Test-before-prod gate is visual only                                                           | CRITICAL                  |        |
|    | The DSD-31718 incident can recur today — the blocked card is an alert, not a lock              |                           |        |
| A2 | DevOps priority list is not a real entity                                                      | CRITICAL                  |        |
|    | Upgrades can be scheduled without Elias/Martin ever seeing or acknowledging them               |                           |        |
| A3 | Complex upgrade checklist not enforced                                                         | CRITICAL                  |        |
|    | High-risk upgrades (<8.17, WildFly 8, on-premise, Java) get a badge but no mandatory step gate |                           |        |
| A4 | Annual upgrade limit not enforced                                                              | HIGH                      |        |

A customer at their contractual limit can be scheduled for another upgrade with no block or warning

|                       |                                                                                                                                                                                                                          |          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| A5                    | <b>Calendar slot conflict detection absent</b><br>Two customers can be silently double-booked into the same DevOps window                                                                                                | HIGH     |
| A6                    | <b>IP/Firewall extended checklist is a flag</b><br>The  badge appears but the step-by-step firewall/integration checklist lives nowhere | HIGH     |
| A7 <small>NEW</small> | <b>Elias rule not enforced — below 8.23 always requires combined upgrade + migration</b><br>A customer below v8.23 can be scheduled for an upgrade-only slot, which will fail at execution time                          | CRITICAL |
| A8 <small>NEW</small> | <b>Out-of-hours slot does not trigger Complex upgrade reclassification</b><br>A Small upgrade booked into a 02:00 UTC window stays classified as Simple — the type and the slot contradict each other silently           | HIGH     |

|          |                                                                                                                                                          |          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>B</b> | <b>Missing Intelligence</b> <small>RISK ACCUMULATES SILENTLY</small>                                                                                     | 6 gaps   |
| B1       | <b>No condition-triggered proactive surfaces</b><br>SLA nearing breach, renewal in 30 days, no contact in 14 days — all invisible until manually checked | CRITICAL |
| B2       | <b>Health score is opaque</b><br>A score of 24 tells a CSM nothing actionable — no formula, no component breakdown, no drill-down                        | HIGH     |
| B3       | <b>No renewal alert workflow</b><br>Renewal dates are field values, not triggers — no automated flag reaches the CSM until it is too late                | HIGH     |
| B4       | <b>Sentiment is a snapshot, not a trend</b><br>A customer deteriorating from Happy → Neutral → Frustrated over 6 weeks is invisible                      | HIGH     |
| B5       | <b>No version → feature/fix matrix</b><br>Cannot articulate to a customer what specific problems upgrading would resolve                                 | MEDIUM   |
| B6       | <b>No fleet-wide upgrade velocity trend</b><br>Stagnation in upgrade cadence across the fleet is invisible until it becomes a version-debt crisis        | MEDIUM   |
| <b>C</b> | <b>Missing Workflows</b> <small>INSTITUTIONAL KNOWLEDGE LOST ON EVERY HANDOVER</small>                                                                   | 6 gaps   |
| C1       | <b>No structured communication log</b>                                                                                                                   | HIGH     |

Customer interaction history lives in personal email — handovers lose context, no audit trail

|        |                                                                                                                                                                |        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| C2     | <b>No proactive outreach tracking</b><br>Releases view identifies who to contact but provides no way to record contact, response, or follow-up                 | HIGH   |
| C3     | <b>No renewal prep workflow</b><br>Renewal is a date, not a process — no checklist, no health-check trigger, no stakeholder alignment steps                    | HIGH   |
| C4     | <b>No escalation path</b><br>A "Chase Dev" button with no owner, no escalation SLA, no status tracking beyond the button press                                 | HIGH   |
| C5     | <b>Out-of-hours scheduling not modelled</b><br>Complex upgrades requiring 02:00 UTC windows cannot be represented or planned in the calendar                   | MEDIUM |
| C6 NEW | <b>No outbound notification channel</b><br>Alerts and escalations surface only inside the tool — they reach no one who is not already looking at the dashboard | HIGH   |

D

**Missing Views** TWO USER GROUPS ENTIRELY UNSERVED

5 gaps

|        |                                                                                                                                                        |          |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| D1     | <b>Tool does not scale beyond 4 customers</b><br>No pagination, no bulk actions, no working import — unusable against 1,000+ real customers            | CRITICAL |
| D2     | <b>No CSM-facing view</b><br>Six named CSMs have no view of their own accounts — they must navigate an L2-centric tool or stay out of it               | HIGH     |
| D3     | <b>No DevOps capacity view</b><br>Elias and Martin's combined workload and ceiling are invisible — upgrade planning is done blind                      | HIGH     |
| D4     | <b>No global search</b><br>A DSD ticket number, customer name, or version string cannot be searched across all views                                   | MEDIUM   |
| D5 NEW | <b>No live infrastructure health check</b><br>Stored version in the DB may diverge from what is actually running on the tenant — no flag, no detection | MEDIUM   |

E

**Data Model Gaps** BLOCKS CORRECT IMPLEMENTATION OF A–D

4 gaps

|    |                                    |      |
|----|------------------------------------|------|
| E1 | <b>Contacts is a single string</b> | HIGH |
|----|------------------------------------|------|

No distinction between primary/technical/billing/emergency contacts; wrong person reached in an incident

|    |                                                                                                                                                 |        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| E2 | <b>SLA not typed by case type</b><br>Defect vs Support vs Upgrade may have different SLA targets — the breach timer may be measuring wrong      | HIGH   |
| E3 | <b>API field is a boolean</b><br><code>api: true</code> doesn't capture which webhooks exist or what needs reconfiguring during migration       | MEDIUM |
| E4 | <b>On-premise inferred from WildFly 8 flag</b><br>A customer running on-premise without WildFly 8 is silently misclassified as a Simple upgrade | MEDIUM |

BUILD PLAN — 7 PHASES

0

**Repository & Docker Foundation**

~2 days · prerequisite for all phases

Set up the full stack from scratch so that every subsequent phase has a working local environment to build against. No features yet — just a running, connected system.

**docker-compose.yml** — four services: db (PostgreSQL 16), api (FastAPI), web (Vue 3 + Vite), nginx (reverse proxy)

**Persistent volume** for PostgreSQL — data survives container restarts

**Alembic initialised** — migration toolchain ready, baseline revision committed

**FastAPI skeleton** — health endpoint, CORS, structured logging, APScheduler initialised, `.env` config pattern

**Vue 3 skeleton** — Pinia store, Vue Router, Axios client pointed at API, design tokens ported to CSS variables

**Hot-reload in dev mode** — `docker compose up` starts everything; file changes reload immediately

1

**Data Model & Core API**

~5 days · addresses E1–E4, enables D1

E1E2E3E4D1 (partial)

Define the full PostgreSQL schema with Alembic migrations. Build the CRUD FastAPI endpoints. Stand up a paginated, filterable customer list in Vue that can handle the full 1,000+ customer base. Two schema elements added here unblock later phases: the `WildFly 8` field (needed for upgrade type logic) and the minimal users table (needed for auth E4s in Phase 2).

later phases. the `wf33` field (needed for upgrade-type logic) and the minimal `users` table (needed for auth FRS in Phase 2).

**users** — minimal table added early: `id`, `name`, `email`, `role` enum (L2/CSM/DevOps/Admin). No full auth yet — just enough to create FK references in subsequent phases.

**customers** — all fields from the prototype plus explicit `on_premise` flag (E4), `wf8` boolean, `wf33` boolean (both modelled; WF33 is the migration target), `sla_tier` FK

**contacts** — structured table: `customer_id`, `name`, `role` (primary/technical/billing/emergency), `email`, `phone` (E1)

**sla\_definitions** — per tier × per case type (Defect/Support/Upgrade) response and resolution targets (E2)

**webhooks** — endpoint, event types, auth method, `migration_tested` flag, linked to customer (E3)

**cases, upgrades, migration\_stages, training\_sessions, training\_gaps, timeline\_events, notes** — all entities from the prototype, now persisted

**Customer list endpoint** — paginated, filtered by tier/CSM/infra/version/health/product, full-text search on name and DSD refs (D1)

**CSV import endpoint** — accepts the Google Sheet export; maps columns, creates customers in bulk, logs conflicts

2

Process Enforcement + Jira Sync

~9 days · addresses A1–A8 · the reason this rebuild exists

- A1
- A2
- A3
- A4
- A5
- A6
- A7
- A8

Every rule that currently lives in team memory becomes a database constraint and API rejection — not just a visual warning. A request that violates a rule is refused by the system, not flagged and ignored. Jira polling replaces manual case entry. Sedna Ops becomes the authoritative scheduling calendar.

**Test-before-prod gate (A1)** — API refuses to advance a PROD upgrade past DevOps Approval if the linked TEST upgrade's status is not `verified`. Resolves the DSD-31718 class of incident.

**DevOps self-assignment pool (A2)** — `devops_queue` table. Asaph submits; either Elias or Martin claims any unclaimed item from their queue view. An upgrade cannot move to Customer Confirmed until its queue entry has an owner and acknowledgment timestamp. Neither engineer is pre-assigned by default — they choose.

**Complex upgrade auto-detection + checklist (A3)** — API evaluates criteria (`version < 8.17`, `on_premise = true`, `wf8 = true`, Java components) and assigns upgrade type. Complex type requires `checklist_items` completion before proceeding.

**Annual upgrade limit (A4)** — API returns 409 when `upgrades_used ≥ upgrades_limit`. Warning at 80%. Manager

- override endpoint with audit log.
- Slot conflict detection (A5)** — upgrade\_slots table with unique constraint on (slot\_datetime, devops\_assignee). Double-booking returns 409.
- IP/FW extended checklist (A6)** — ip\_fw flag triggers mandatory checklist set for migration. Cannot advance past Customer Contacted without firewall checklist completion.
- Elias rule enforcement (A7)** — API rejects any upgrade-only request where customer version < 8.23. The system enforces combined upgrade + migration for this cohort; no manual override without admin approval and audit log. Configurable threshold in admin settings.
- Out-of-hours → Complex reclassification (A8)** — when a slot is assigned an out-of-hours time window, the API checks upgrade type. If classified as Simple or Standard, it surfaces a mandatory reclassification prompt. Confirmation as Complex required before slot is saved.
- Jira polling (A2 enabler)** — APScheduler background task polls Jira every 5 minutes: JQL project = DSD AND updated >= -5m ORDER BY updated DESC, Jira API key auth. New cases upserted into the cases table. Sedna Ops does not modify Jira — polling is read-only.
- Authoritative calendar** — Sedna Ops becomes the scheduling source of truth. Google Calendar dependency eliminated. Slots created here; customer confirmation still recorded in Jira by Asaph; Asaph records the confirmation in Sedna Ops with one click.
- AI use case 1 — Case classification (Haiku 4.5, automated)** — on every Jira sync, new cases are sent to Haiku for classification (case type, priority signal, upgrade relevance). Results stored with input\_hash; classification is skipped on re-poll if case text unchanged. ~\$0.77/month at 25 new cases/day.

3

Intelligence Layer + Notifications

~8 days · addresses B1–B6, C6

- B1
- B2
- B3
- B4
- B5
- B6
- C6

- Derive signal from the data that is now persistent. The app stops being a record-keeper and starts surfacing what needs attention before you would otherwise notice it. Alerts reach people outside the tool via Slack webhook — not just the dashboard.
- Proactive alert engine (B1)** — APScheduler evaluates conditions every 15 minutes: SLA within 20% of breach, renewal within 90/60/30 days, no contact in 14+ days, migration stalled for 7+ days. Alerts surface in Triage Queue and Alerts panel.
  - Health score engine (B2)** — formula exposed as a documented API endpoint. Weights: version age 25%, open critical

**Health score engine (B2)** — formula exposed as a documented API endpoint. Weighed: version age 20%, open critical cases 25%, training gap recurrence 15%, days since upgrade 15%, infra type 10%, sentiment 10%. Drill-down shows each component's contribution and what would improve the score.

**Renewal alert workflow (B3)** — at 90/60/30 days before renewal: alert created, assigned to CSM, marked urgent if health < 60. Includes pre-populated renewal prep checklist.

**Sentiment history (B4)** — sentiment\_log table with trend. Timeline tab shows direction. Customer 360 shows current value + trend arrow.

**Version feature matrix (B5)** — release\_features table. Customer 360 Upgrades tab shows "you would gain X fixes by upgrading from your current version."

**Upgrade velocity trend (B6)** — monthly aggregates in a materialised view. 12-month fleet cadence chart in Trends view.

**Slack outbound webhook (C6)** — a single Slack webhook URL (configured by Asaph as admin) receives alert notifications: SLA breach, renewal urgency, blocked upgrade, escalation created. Formatted as Slack block messages. Reused by C4 (escalations) in Phase 4. C3 waits for B3 to be complete before the renewal prep workflow is built on top of it.

**AI use case 2 — Training gap detection (Haiku 4.5, automated)** — when a case is marked resolved, Haiku reads the case summary and resolution notes and identifies any training gap pattern. Tags the case with gap category and affected module. Results stored; no re-classification if data unchanged. ~\$0.20/month.

4

Workflow Layer

~6 days · addresses C1–C5

- C1
- C2
- C3
- C4
- C5

Give every business process a home in the tool. What currently lives in email inboxes, Slack threads, and personal memory becomes a structured record that survives the person who created it.

**Communication log (C1)** — comms\_log table: customer\_id, direction (inbound/outbound), channel (email/call/Slack/meeting), subject, summary, sent\_by, timestamp, linked\_case\_ref. New tab in Customer 360. Exportable per customer.

**Proactive outreach tracking (C2)** — Release impact list generates an outreach campaign: each affected customer gets an outreach\_record with status (pending/sent/responded/resolved), linked comms log entries, and follow-up due date.

**Renewal prep workflow (C3)** — built on top of B3 renewal alerts. At the 90-day alert, a renewal\_prep record is created with a templated checklist (health check, defects acknowledged, CSM call scheduled, commercial terms reviewed, upgrade plan agreed). Visible to both L2 and CSM.

**Escalation path (C4)** — escalations table: linked case, reason, target team (Dev/DevOps/Management), owner, SLA hours, status, resolution notes. "Chase Dev" creates a real escalation record. Slack notification sent via the webhook from Phase 3. Overdue escalations surface in Triage.

**Out-of-hours scheduling (C5)** — slots have a time field (not just date) and an out\_of\_hours flag. Complex upgrade booking UI allows any UTC time. Display converts to customer timezone.

**AI use case 3 — Draft communications (Sonnet 5, on-demand)** — "Draft Confirm" and "Draft Follow-Up" buttons in the comms log panel. Sends the customer profile snapshot + recent case context to Sonnet. Returns an editable draft. User clicks; user edits; user sends. No automatic sending. ~\$1.05/month at 5 drafts/day.

5

Role Views, Global Search & AI Synthesis

~8 days · addresses D2–D4

D2 D3 D4

The tool currently serves one role. Three distinct audiences — L2 (Asaph), CSMs, and DevOps (Elias/Martin) — get views tailored to what they actually need to know and do. On-demand AI synthesis surfaces here for the two customer-facing roles.

**CSM view (D2)** — a separate route (/csm) showing only the accounts assigned to the logged-in CSM. Columns tuned to commercial concerns: health score, renewal date, open cases, migration status, last contact date. Renewal prep checklists accessible here. No L2 pipeline detail.

**DevOps capacity view (D3)** — queue of all items awaiting Elias/Martin: unclaimed items (either engineer claims any), active migrations by assignee, estimated hours per item, weekly capacity ceiling (configurable by Asaph as admin). Colour-coded by assignee load. Engineers can claim and acknowledge items from this view — self-assignment pool model, not pre-assigned by Asaph.

**Global search (D4)** — PostgreSQL full-text index across customers (name, region), cases (DSD ref, title), upgrade history, migration notes, comms log. Results grouped by type. Persistent search bar in top navigation with keyboard shortcut.

**Authentication** — role-based access using the users table from Phase 1: L2 sees everything; CSM sees their accounts; DevOps sees capacity view and queue. Simple role check — not a full auth system yet.

**AI use case 4 — Customer situation summary (Sonnet 5, on-demand)** — "Summarise for Gisele" button on the Customer 360 panel. Sonnet reads the full customer context (cases, health, sentiment, comms log, upgrade history) and produces a 5-8 sentence narrative. Cached by input\_hash — regenerated only when underlying data changes. ~\$0.99/month at 3 summaries/day.

**AI use case 5 — Proactive action recommendation (Sonnet 5, on-demand)** — "What should I do next?" button per customer. Sonnet evaluates the full customer state and recommends the one highest-leverage action Asaph or the CSM should take this week. ~\$0.72/month at 2 recommendations/day.

6

Live Infrastructure, Reporting & Fleet Intelligence

~6 days · addresses D5, snapshot, export

D5 Snapshot Export

Replace the mocked infrastructure check with real API calls to the Dataloy version endpoint. Wire the Snapshot view to real data. The weekly fleet intelligence brief replaces Asaph's manual Slack summaries to Gisele with AI-generated strategic synthesis.

**Live version check (D5)** — FastAPI calls {tenant}.dataloy.com/ws/rest/dataloy/info and returns real version, environment type, and feature flags. Results cached for 10 minutes. Discrepancies between stored version and live version surface as a flag on the customer record.

**Real Snapshot view** — end-of-day summary generated from live database state: SLA breaches today, upgrades completed today, items blocked, follow-ups required tomorrow. Copyable as plain text for Slack or email.

**Management export** — Trends view "Export" generates a structured summary: revenue at risk by category, migration programme progress vs target, fleet version distribution, CSM workload overview. Formatted for a non-technical reader.

**Performance hardening** — database indexes for query patterns identified in Phase 1–5. Connection pooling (asyncpg). Verify paginated list performance at 1,000+ rows.

**AI use case 6 — Weekly fleet intelligence brief (Sonnet 5, scheduled)** — APScheduler fires one Sonnet call per week. Input: aggregated DB stats (case volume by product area, version distribution histogram, training gap frequency table, top 10 customers by health score decline, renewal pipeline, upgrade velocity). Sonnet's system prompt contains cached Sedna/Dataloy domain context. Output: 5-8 sentences surfacing the non-obvious pattern, risk, or opportunity across the full fleet — the kind of cross-customer insight no rule-based system can produce. Delivered as a pinned brief on the Dashboard; optionally pushed to Slack via the webhook from Phase 3. ~\$0.09/month.

| USE CASE                        | MODEL                                                                                                                                                                                                                                            | TRIGGER                                                      | PHASE                                                                                                                                                                                           | EST. COST/MO |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Case classification             | Haiku 4.5                                                                                                                                                                                                                                        | Automated — fires on every Jira sync for new/changed cases   | 2                                                                                                                                                                                               | ~\$0.77      |
| Training gap detection          | Haiku 4.5                                                                                                                                                                                                                                        | Automated — fires when a case is resolved                    | 3                                                                                                                                                                                               | ~\$0.20      |
| Draft communications            | Sonnet 5                                                                                                                                                                                                                                         | On-demand — user clicks "Draft Confirm" or "Draft Follow-Up" | 4                                                                                                                                                                                               | ~\$1.05      |
| Customer situation summary      | Sonnet 5                                                                                                                                                                                                                                         | On-demand — user clicks "Summarise for Gisele"               | 5                                                                                                                                                                                               | ~\$0.99      |
| Proactive action recommendation | Sonnet 5                                                                                                                                                                                                                                         | On-demand — user clicks "What should I do next?"             | 5                                                                                                                                                                                               | ~\$0.72      |
| Weekly fleet intelligence brief | Sonnet 5                                                                                                                                                                                                                                         | Scheduled — one call per week, aggregated DB stats as input  | 6                                                                                                                                                                                               | ~\$0.09      |
| → Two-tier model split.         | Haiku 4.5 (\$1.00/\$5.00 per 1M tokens) handles all automated/batch tasks. Sonnet 5 (\$3.00/\$15.00) is reserved for human-triggered synthesis that demands nuanced judgment. Never use Sonnet for classification that Haiku handles adequately. |                                                              |                                                                                                                                                                                                 |              |
| → Prompt caching.               | The Sedna/Dataloy domain context (upgrade rules, tier definitions, role descriptions) goes in a cached system prompt. Cache hits are billed at 10% of the normal input rate — saves \$0.50–1.00/month across all calls.                          |                                                              |                                                                                                                                                                                                 |              |
| → DB-level result caching.      | Every AI output is stored with an                                                                                                                                                                                                                | input_hash and generated_at                                  | timestamp. The API skips Claude and returns the cached result if the underlying data has not changed since the last call. Summaries and recommendations are not regenerated on every page load. |              |
| → Compact context objects.      | Each task type receives only the fields it actually needs — not the full customer object. Minimises input tokens per call without losing signal.                                                                                                 |                                                              |                                                                                                                                                                                                 |              |

**Build sequence rationale:** Phase 0 is the foundation every other phase depends on. Phase 1 must come before Phase 2 because enforcement is only meaningful when the data model can represent what is being enforced — including the WF33 field and minimal

users table, both needed by subsequent phases. Phase 2 comes before intelligence (Phase 3) because surfacing proactive alerts about process violations requires those violations to actually be detectable. C3 (renewal prep workflow) in Phase 4 depends on B3 (renewal alerts) in Phase 3. The AI use cases in Phases 2 and 3 are automated and light (Haiku); the heavier Sonnet synthesis in Phases 4 and 5 is on-demand only, so cost scales with user behaviour, not background task frequency. The fleet intelligence brief in Phase 6 runs on real data; building it last ensures there is enough history in the system for the synthesis to be meaningful.